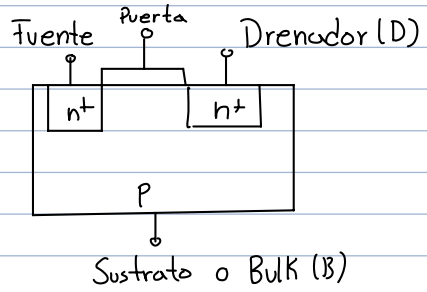
Transistores de efecto de campo:

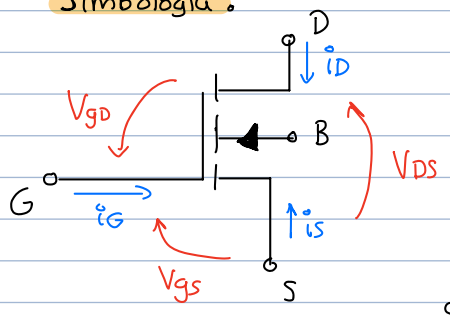
- Principio de funcionamiento en contínua:

- Mosfet de acumulación:

Canal n:



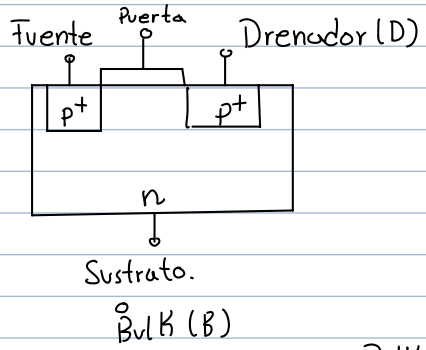
Simbología:

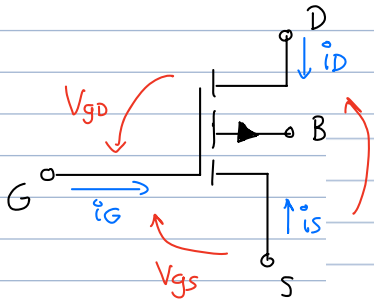


$i_g = 0$; $i_D = -i_S$

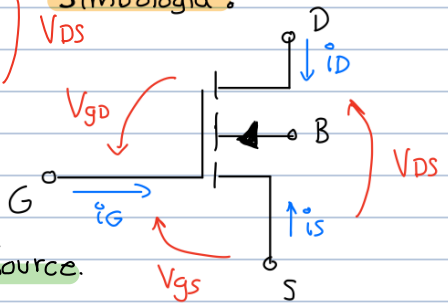
$V_{GS} = V_{DS} + V_{GD}$.

Canal p:





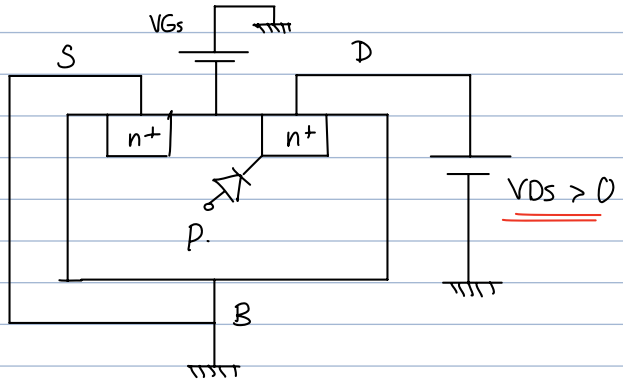
Simbología:



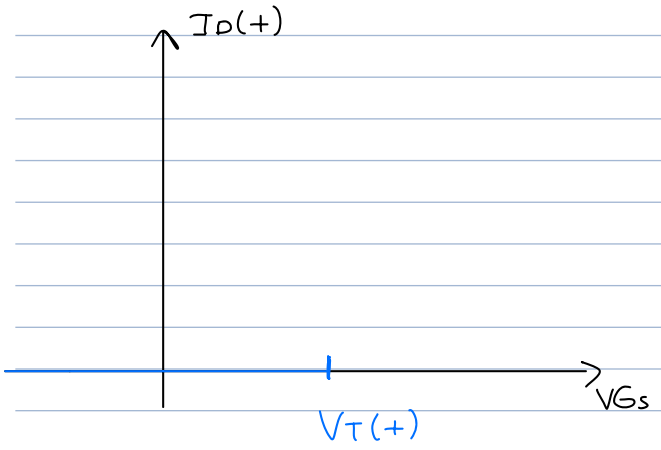
Bulk conectado generalmente a Source.

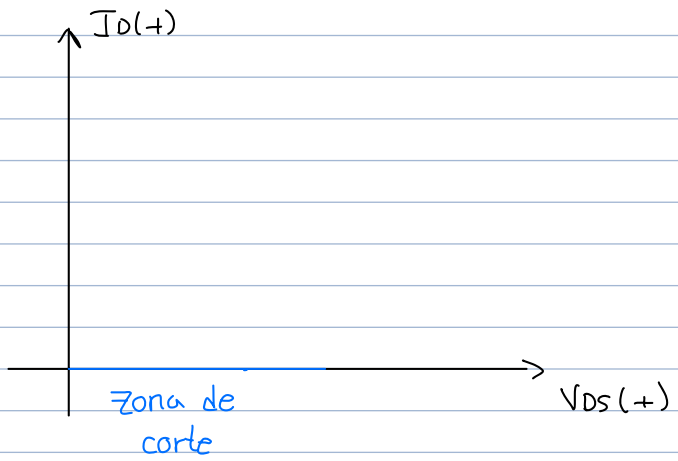
- Estudio cualitativo:

Transistor de acumulación canal n: Zona de corte.



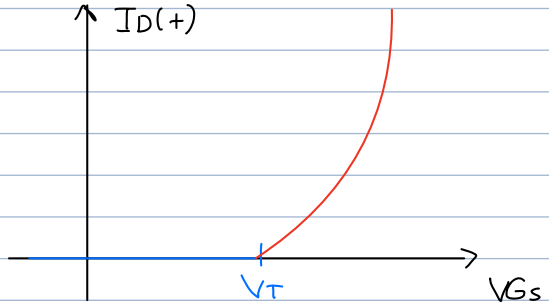
Se forma un diódo, tiene que funcionar en inversa. Para funcionar en inversa en un transistor de acumulacion canal n, $V_{DS} > 0$.

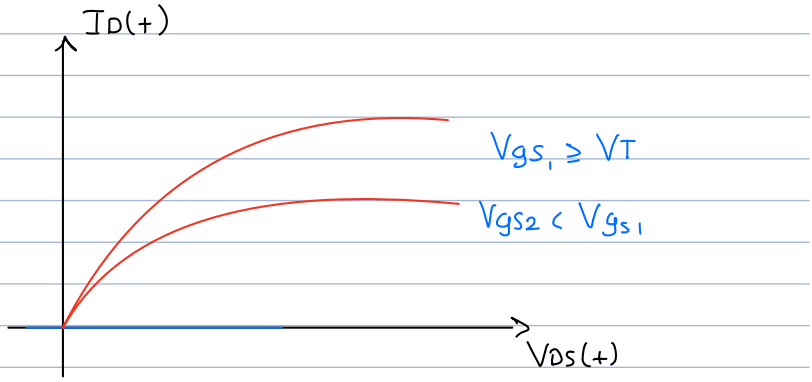




Condición **Corte** ⟶ $V_{GS} \leq V_T$

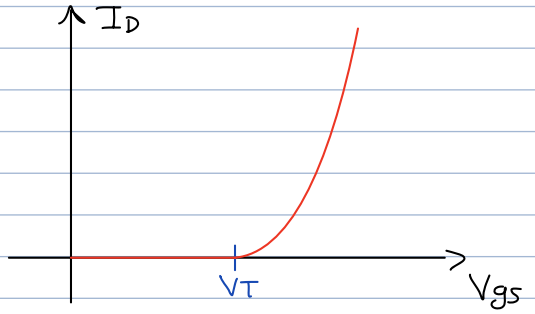
Zona ohmica:

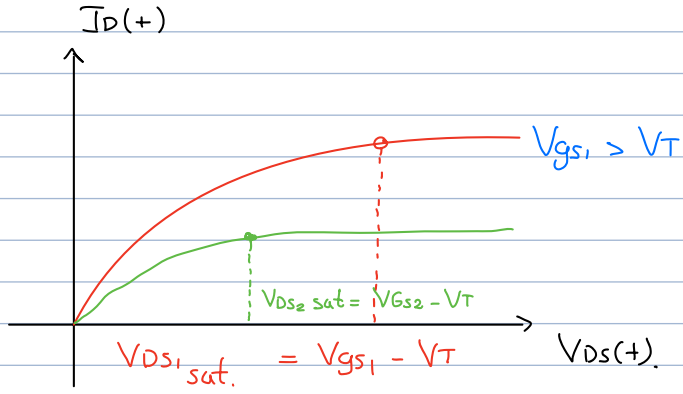




Condición **Ohmica**: $\begin{cases} V_{GS} \geq V_T \\ I_D \sim K \cdot V_{DS} > 0 \end{cases}$

Zona saturación:





La diferencia entre la **zona ohmica** y la **zona de saturación**.

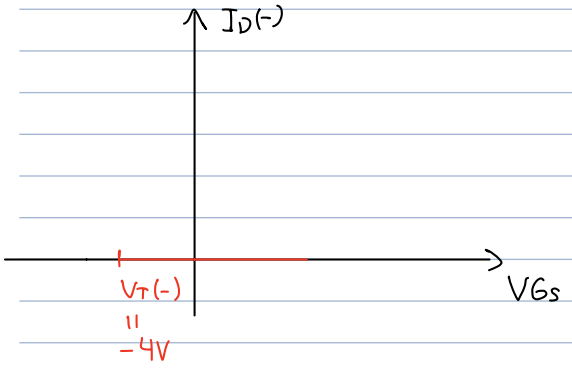
Ohmica: $\begin{cases} V_{GS} \geq V_T \\ V_{DS} \leq V_{GS} - V_T \end{cases}$

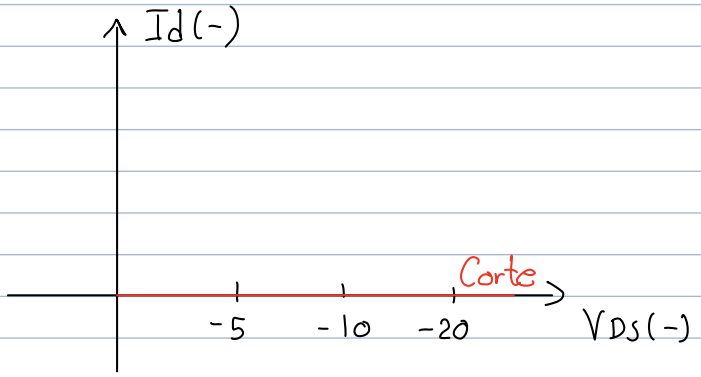
Saturación: $\begin{cases} V_{GS} \geq V_T \\ V_{DS} \geq V_{GS} - V_T \end{cases}$

$V_{DS} = V_{GS} - V_T$

Transistor de acumulación p :

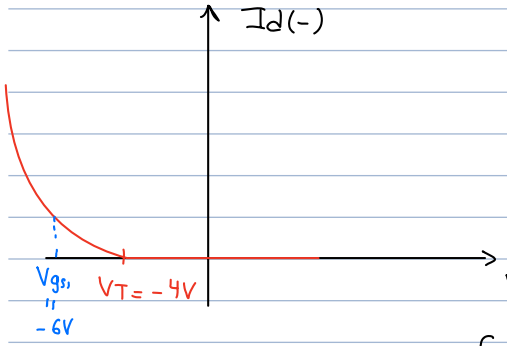
Zona de corte :

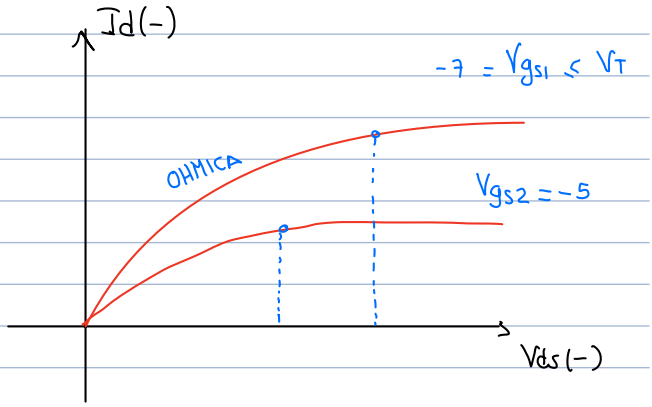




Condición corte : $V_{gs} \geq V_T$

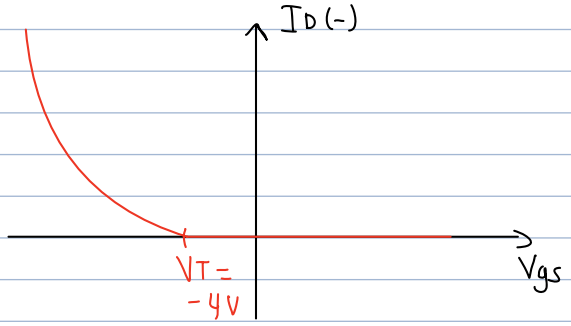
Zona ohmica :

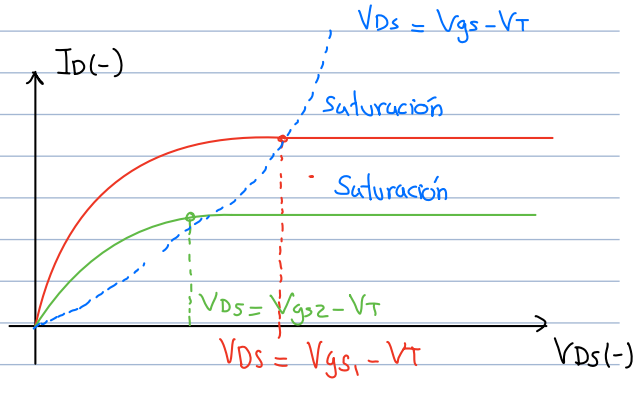




Condición ohmica : $\begin{cases} V_{gs} < V_T \\ V_{DS} \geq V_{gs} - V_T \end{cases}$

Zona saturación :





Condición saturación : $\begin{cases} V_{gs} < V_T \\ V_{DS} < V_{gs} - V_T \end{cases}$

- Análisis cuantitativo:

Zona ohmica: $i_D = K \cdot \left((V_{gs} - V_T) \cdot V_{ds} - \frac{V_{ds}^2}{2} \right)$ → Se linealiza.

El parámetro K viene medido en mA/V^2

Zona saturación:

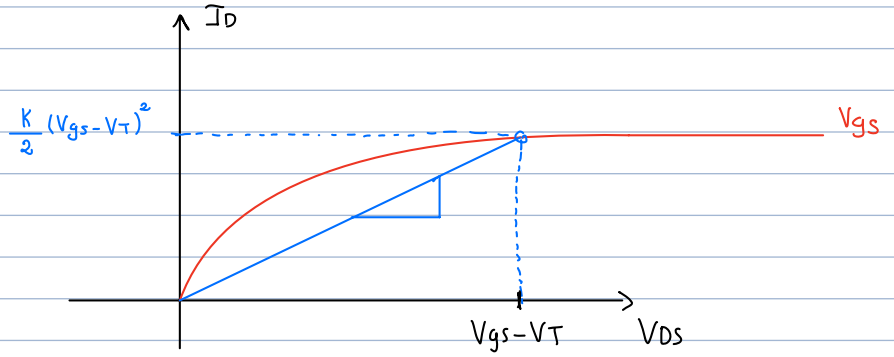
$$i_{Dsat} = \frac{K}{2} (V_{gs} - V_T)^2$$

En un canal n: $K > 0$, $V_T > 0$, $i_D > 0$, $V_{DS} > 0$

canal p: $K < 0$, $V_T < 0$, $i_D < 0$, $V_{DS} < 0$.

- Curvas características de salida aproximadas:

Canal n y canal p:

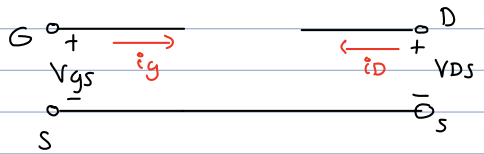


Linealizamos las curvas, en este caso sustituimos por una resistencia.

$$R_{DSon} = \frac{V_{gs} - V_T}{\frac{K}{2} \cdot (V_{gs} - V_T)^2} = \frac{1}{\frac{K}{2} \cdot (V_{gs} - V_T)}$$

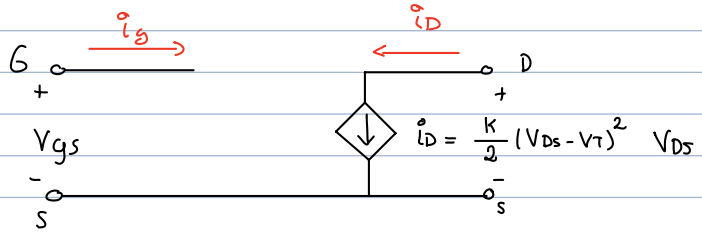
Circuitos equivalentes canal n:

Zona corte:



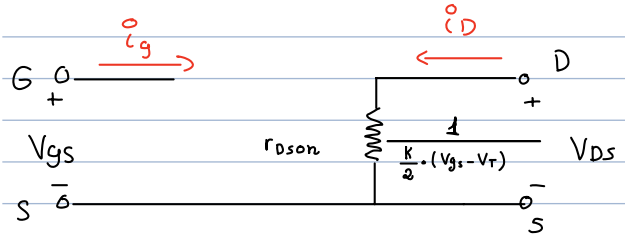
$V_{gs} \leq V_T$

Zona saturación:



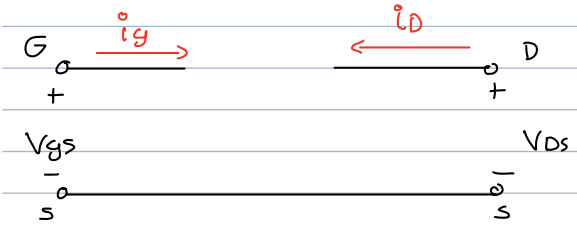
$V_{gs} > V_T$; $V_{DS} \geq V_{gs} - V_T$

Zona ohmica:



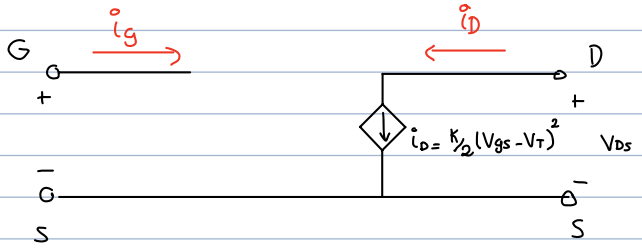
Circuito equivalente canal p:

Zona de corte:



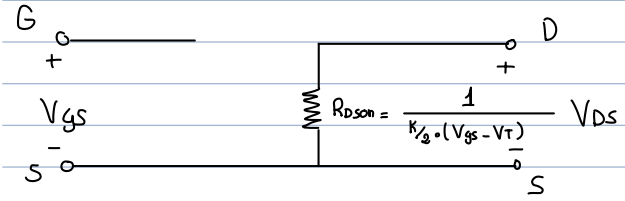
$V_{GS} \geq V_T$

Zona de saturación:



$V_{GS} \leq V_T$; $V_{DS} \leq V_{GS} - V_T$

Zona ohmica:



$V_{GS} < V_T$; $V_{DS} \geq V_{GS} - V_T$

- Análisis en continua: determinación de zona de trabajo:

1º- Hipótesis zona de trabajo

2º- Sustituir MOSFET por su modelo equivalente

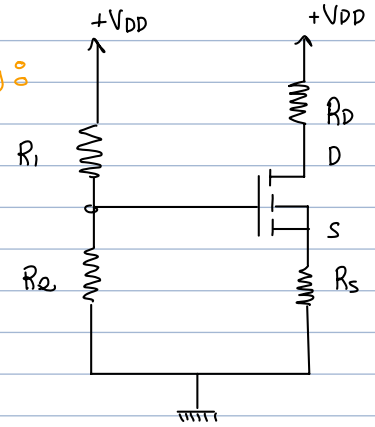
- Saturación → Acumulación → Canal n: $V_{GS} > V_T$, $V_{DS} \geq V_{GS} - V_T$
  - → Canal p: $V_{GS} < V_T$, $V_{DS} \leq V_{GS} - V_T$.
  - → Deplexión → Canal n: $V_{GS} > V_T$, $V_{DS} \geq V_{GS} - V_T$
    - → Canal p: $V_{GS} < V_T$, $V_{DS} \leq V_{GS} - V_T$.

- Ohmica → Acumulación → Canal n: $V_{GS} > V_T$; $V_{DS} \leq V_{GS} - V_T$.
  - → Canal p: $V_{GS} < V_T$; $V_{DS} \geq V_{GS} - V_T$.
  - → Deplexión → Canal n: $V_{GS} > V_T$; $V_{DS} \leq V_{GS} - V_T$
    - → Canal p: $V_{GS} < V_T$; $V_{DS} \geq V_{GS} - V_T$.
- Corte → Acumulación → Canal n: $V_{GS} \leq V_T$
  - Canal p: $V_{GS} \geq V_T$
  - → Deplexión → Canal n: $V_{GS} \leq V_T$
    - Canal p: $V_{GS} \geq V_T$

3° Resolver sistemas de ecuaciones:

4° Si hipotesis correcta. OK. Si no, nueva hipótesis.

Ej:

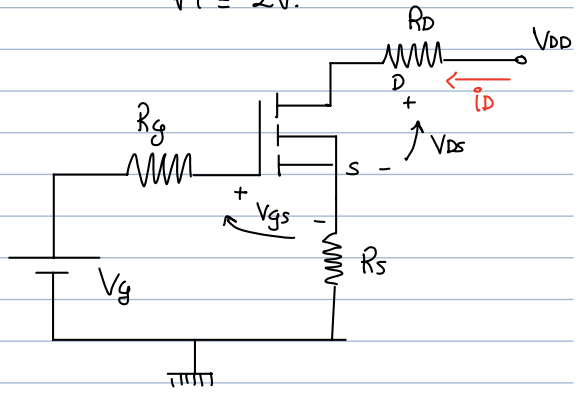


Eq. Thevenin →

$$V_g = V_{DD} \frac{R_2}{R_1 + R_2}$$

$$R_g = \frac{R_1 R_2}{R_1 + R_2}$$

Datos: $R_1 = 3M\Omega$; $R_2 = 1M\Omega$
$R_D = 4{,}7K\Omega$ $R_S = 2{,}7K\Omega$
$V_{DD} = 20V$ $K/2 = 1mA/V^2$
$V_T = 2V$.



Hipótesis: Mosfet en saturación:

$$I_D = I_{Dsat} = K/2\,(V_{GS} - V_T)^2$$

$$\left.\begin{aligned} V_g &= V_{GS} + R_S I_D \\ V_{DD} &= I_D(R_D + R_S) + V_{DS} \\ I_D &= K/2 \cdot (V_{GS} - V_T)^2 \end{aligned}\right\}$$

$$V_{DD} = I_D(R_D + R_S) + V_{DS}$$

$$\frac{V_g - V_{GS}}{R_S} = \frac{K}{2} \cdot (V_{GS} - V_T)^2$$

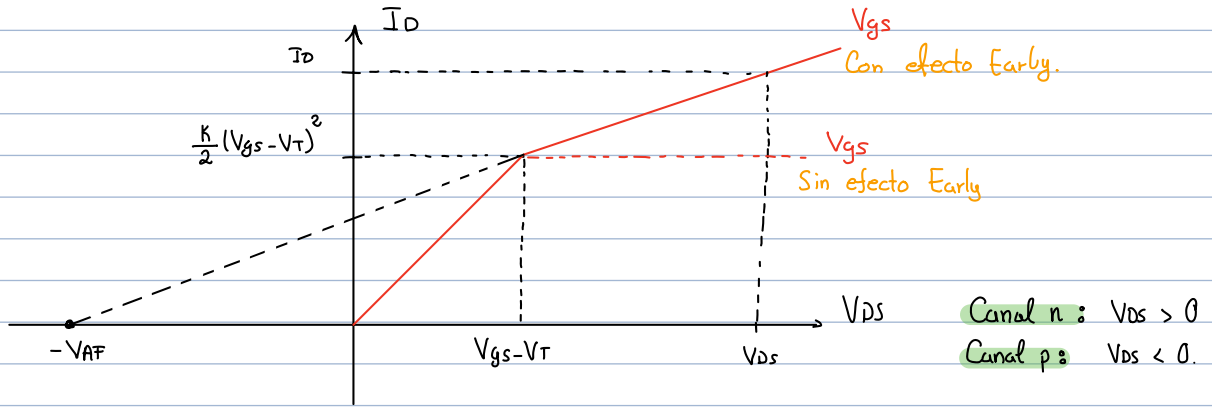
$$\frac{V_g - V_{GS}}{R_S} = \frac{K}{2}(V_{GS} - V_T)^2 \Rightarrow V_{GS}^2 - 3{,}6\,V_{GS} + 2{,}15 = 0 \Rightarrow V_{GS1} = 2{,}89V$$

$$V_{GS2} = 0{,}74V \not< V_T$$

$$I_D = \frac{K}{2}(V_{GS} - V_T)^2 = 0{,}78\ mA \quad ; \quad V_{DS} = V_{DD} - I_D(R_D + R_S) = 14{,}2V$$

$V_{DS} > V_{GS} - V_T \Rightarrow$ Mosfet en saturación

- Efectos de segundo orden : Efecto Early :



$$i_D = \frac{K}{2}(V_{GS} - V_T)^2 + \frac{V_{DS}}{r_o} \qquad \text{donde} \qquad r_o = \frac{V_{AF}}{\frac{K}{2}(V_{GS} - V_T)^2}$$

